

Instructor: Date:

Learner: Regional Branch:

Program: Center:

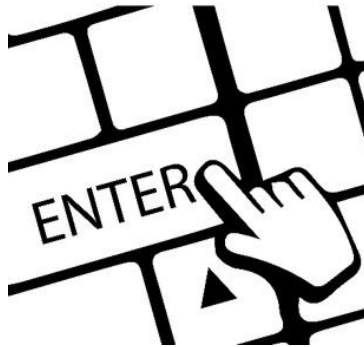
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Let's Start!

Deepening the Concept of Functional Requirements



Learning Outcome:

Make the coding of the system modules and the main program, starting from using the chosen programming language, based on design specifications.
Exchanges information about requirements types and their characteristics.

Can Do Statements:

- I can talk about the advantages and disadvantages of using functional requirements.
- I can tell how functional requirements looks like in a use case and in a user story.

Ask your partner:

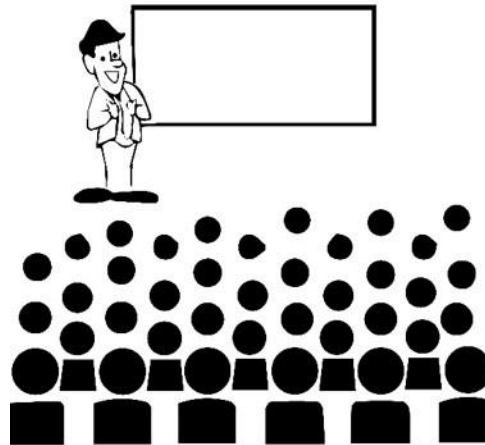
- How can you instruct people into writing good functional requirements? Share some of your ideas.
- Do you consider it is easy to identify your own mistakes when it comes to writing functional requirements? Why? Why not?





Let's Explore!

Prepare your answers!



Instructions: Read the text below.

- Margarita:** Professor Gonzalez, who is a prominent figure in the field of programming, visits us this afternoon. He will discuss important aspects of the functional requirements, and I'm going to prepare as many questions as possible. I hope he addresses the topic of functional requirements today.
- Carlos:** I think I will ask him about the advantages and disadvantages of the functional requirements.
- Margarita:** You don't have to ask Mr. Gonzalez! I can give you the answer right now. On the one hand, the advantages of functional requirements are really easy to understand. When you write a requirement that has good syntax and provides the complete and concise idea of what you want the system to perform, it is going to be reflected on the final product and the way it is achieved. All the following steps that are coming after the requirement is written will flow smoothly, and there will be no place for misinterpretations. On the other hand, the disadvantages of functional requirements will show up only when they are not necessary, concise, achievable, unambiguous, complete nor verifiable. They will impede the requirement to convert into a fully operational system or application.
- Carlos:** Wow. I'll ask him how to use case and user stories look like, then.
- Margarita:** Well, you don't have to ask that either. The functional requirements in use cases look like a series of different actions that describe the steps of the interactions between the system and the role. Now, the user stories portray a detailed description of the uses of the system which will guide and teach the user how to use it.
- Carlos:** Shoot, I don't know what I am going to ask him now.
- Margarita:** Think about that later. Now, we have to hurry up, Mr. Gonzalez' conference starts in 30 minutes.



Let's Practice!

Margarita knows almost everything about functional requirements. She needs to understand those words.

Instructions: Match column A with column B.

COLUMN A

1. The grammar structure or sequence of the components in the language
2. To make or say just what is necessary
3. To describe a user need
4. The behavior to form into the system
5. To do an action or task which needs training

COLUMN B

- a. Perform
- b. Use case
- c. User stories
- d. Syntax
- e. Concise



Let's Practice!

Mr. Gonzalez is coming.

Instructions: Select the correct options to complete the sentences.

1. Asking about the advantages and disadvantages of the functional requirements
(Hint: (Good is an irregular adjective.)
a. is not the best question to Mr. Gonzalez.
b. is worst question to Mr. Gonzalez.
c. is the more appropriate question to Mr. Gonzalez.
2. The advantages of functional requirements
(Hint: 'Requirements is plural.)
a. are the easiest to understand.
b. are the easy to understand.
c. is the easiest to understand.
3. Requirements with
(Hint: Use the superlative form of 'good'.)
a. the most achieving syntax and concise ideas of what you want the system to perform are going to be reflected on the final product and the way those are achieved.
b. the least syntax and concise ideas are going to be reflected on the final product and the way those are achieved.
c. the best syntax and concise ideas are going to be reflected on the final product and the way those are achieved.





Let's Practice!

4. The steps that are coming after the requirement is written will be
(Hint: Smooth is a one-syllable adjective.)
 - a. the smoothiest and there will be no place for misinterpretations.
 - b. the smoothest, and there will be no place for misinterpretations.
 - c. the smooth and there will be place for interpretations.
5. The most challenging issue is to
(Hint: No need to use a superlative form.)
 - a. communicate with simple words.
 - b. write difficultest words.
 - c. not communicate difficulter words.



Extension Activity

My Use Case and User Story Poster Design

Instructions: Complete the following steps. Fill in the Self-Assessment.

- a) Look for further information about use case and user story in functional requirements.
- b) Make notes about the main features in each one of them
- c) Design a poster in which you show the advantages and disadvantages of using functional requirements.
- d) Share with your tutor and answer all his questions.

Self-Assessment

Yes

No

1. I look for information about use case and user story in functional requirements.
2. I design a poster about use case and in a user story features.
3. I talk about the advantages and disadvantages of using functional requirements.

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Target Words

- **prominent figure** (noun): a very knowledgeable person; an expert on a specific field
e.g. Professor Gonzalez, who is a prominent figure in the field of programming, visits us this afternoon.
- **smoothly** (adverb): without any trouble or interruption
e.g. All the following steps that are coming after the requirement is written will flow smoothly.
- **misinterpretation** (noun): a wrong and incorrect understanding of a topic, situation, or thing
e.g. There will be no place for misinterpretations.
- **achievable** (adjective): possible to achieve or accomplish
e.g. The disadvantages of functional requirements will show up only when they are not necessary, concise, achievable, unambiguous, complete nor verifiable.
- **impede** (verb): to not allow something to take place or happen
e.g. They will impede the requirement to convert into a fully operational system or application.
- **shoot** (expression): an exclamation form to express annoyance
e.g. Shoot, I don't know what I am going to ask him now.



Learning Points

The Future With the Present Simple

The **simple present** tense does not always refer to the present time. You can also use it to talk about the future.

The **simple present** tense can also have a future connotation as it can refer to events taking place in the near **future**. In order to change its connotation of time, we have to add a time expression right next to the sentence we write in **simple present**. As you are referring to specific events in a specific moment in the **future**, the time expressions that you use have to be as precise as possible; thus, expressions like *tonight at 9 o'clock*, *in half an hour*, or *tomorrow at 6 pm* are needed.

Structure:

Subject + **verb (present simple)** + time expression (precise in the future).

Examples:

*The meeting to check the business goals **takes** place **next Monday**.*

*The supervisors **meet** **tonight**.*





Learning Points

Examples

- *Professor Gonzalez, who is a prominent figure in the field of programming, visits us this afternoon.*
- *Now, we have to hurry up, Mr. Gonzalez' conference starts in 30 minutes.*

Functional Requirements: In a Use Case and an User Story

Functional requirements must contain two very important factors in order to achieve success, use cases and user stories.

USE CASE

A use case is basically a list of a series of actions that describe the interactions and relationship between the developed system and a role, which is basically the actor (human user or other systems) in the achievement of a goal. The way these are written are by sequence steps or flows following the format *(Actor/System) does (action)*.

USER STORY

The user stories show in a short but detailed description of the different uses of the developed system. These explanations are done step-by-step and are meant to guide and teach the user how to use the system. The language that is used in user stories is simple and basic so any user can understand; however, it tends to follow a specific model: *As a (human user or other system) I want (specific action from the system) so that (final achievement)*.